Kroger Post-Pandemic Customer Segmentation

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Introduction

The grocery retail industry landscape has changed greatly in the wake of the pandemic. Specifically, delivery and pick-up services have become more popular and customer buying habits have evolved. At the same time, improvements in data collection and analysis have allowed grocery marketing strategies to become highly individualized.

We worked with 84.51, an analytics firm, to identify customer segmentations for the Kroger Company based on data from 2023. Using clustering techniques, we organized customers into groups, or segments, based on similar characteristics. Kroger takes pride in understanding and serving their customer base. Customer segmentations can inform Kroger's efforts to create effective marketing strategies, develop product offerings, and enhance the customer experience.

Data

The data set is comprised of 32 demographic and sales related attributes for 100,000 Kroger customer households. Sales related attributes are aggregated over a period of one year (2023).

Customer Demographics

Household size		Household Income		Education level		Customer Loyalty	
Single	20%	Low (0-52)	16%	< High	7%	Loyal	73%
Small (2-3)	54%	Medium	63%	High school	49%	Opportunity	16%
Medium	21%	High	21%	Undergraduate	34%	Non-Loyal	6%
Large	5%			Graduate	10%	New Household	5%

Sales Related Attributes

Coupon Frequency: Number of visits a coupon was used.
Yellow Tag Freq.: Number of visits a yellow tag discount was used.
Kroger Sales: Total sales of Kroger brand products.
Non-Kroger Sales: Total sales of non-Kroger brand products.

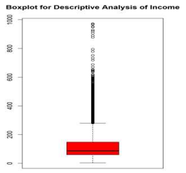
For each shopping mode (In store, Delivery, and Pick-up), the following attributes are recorded:

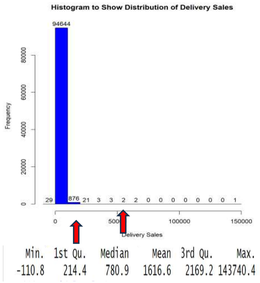
Sales: Total sales.
Units: Total number of items purchased.
Visits: Total number of times the shopping method was used.
Sales per Visit: Average sales for each visit.
Coupon Discount: Total savings from coupons.
Yellow Tag Discount: Total savings from yellow tag discounts.
Total Discount: Total savings of all discounts.

Data Preparation

Before beginning our analysis, we screened the data to identify households or records with extreme values, missing values, or values that didn't make sense. Since only 4.4% of the data was missing, we decided to remove these households from the dataset.

Further, boxplots with fences and histograms were constructed for each attribute to examine anomalies. Both invalid and extreme values were identified and removed.

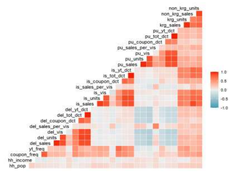




After removing the invalid records and extreme outliers, 94,472 households remained in our dataset.

Multicollinearity

There was a strong presence of multicollinearity in the dataset. To address the issue, we eliminated one of two variables that had a correlation coefficient of 0.95 or higher with each other.



We eliminated the Kroger units, pick-up units, delivery units, pick-up total discount, in store total discount, and delivery total discount variables. We have 26 attributes remaining in the dataset.

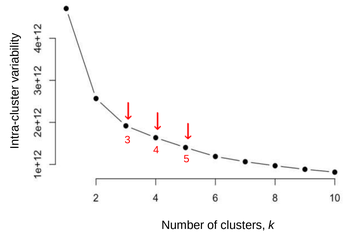
K-means Clustering

K-means is an unsupervised clustering method that groups data points based on their Euclidean distances from the centroids. Given *n* data points, *k* clusters, and *k* arbitrarily selected centroids, the algorithm assigns each data point to a cluster based on the distance to the nearest cluster centroid. Cluster centroids are then recalculated and data points are reassigned based on the new centroids. This process repeats until the centroids no longer move.

We used the intra-cluster variability, the sum of the squares of the distance from each data point to its nearest centroid, as a tool to measure the effectiveness of the clustering. A scree plot shows how the intra-cluster variability varies with the number of clusters, *k*, and can be used to determine the optimal number of clusters.

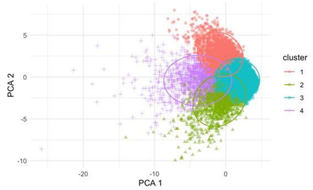
To make the computations more manageable, we worked with a representative sample of 10,000 households.

Scree Plot



After examining 3, 4, and 5 clusters, we decided to move forward with 4 clusters.

Principle Component Analysis (PCA) Plot

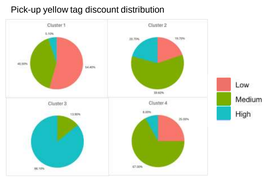


The PCA plot visualizes four clear clusters in our high dimensional data.

Profiling

To profile the clusters, we examined the summary statistics for each attribute (in each cluster). To make the profiling clearer, we recoded quantitative variables into categorical variables. Each quantitative variable was split into 3 levels (high, medium, and low) using the 1st and 3rd quartiles. For each attribute in each cluster, we calculated the percentage of households who fell into each level. These results were then used to classify each cluster.

All				
Attributes	Cluster 1	Cluster 2	Cluster 3	Cluster 4
Pickup Yellow Tag Discount	Low	Medium	High (86.1%)	Medium
Pickup Coupon Discount	Medium	Medium	High (57.7%)	Medium
Pickup Sales per Visit	Medium	Medium	Medium(58.5%)+High (41.5%)	Medium
Pickup Sales	Low	Medium	High (87.7%)	Medium
Pickup Visits	Low	Medium	High (80.4%)	Medium
Instore Yellow Tag Discount	Medium	High (96%)	Medium	Medium
Instore Coupon Discount	Medium	High (77.1%)	Medium	Medium
Instore Sales per Visit	Medium	Medium (52.4%) + High (29.8%)	Medium	Medium
Instore Sales	Medium	High (95.7%)	Medium	Medium
Instore Visits	Medium	High (84.6%)	Medium	Medium
Instore Units	Medium	Medium (56.7%)	Medium	Medium
Delivery Yellow Tag Discount	High(97%)	Medium	Medium	Medium
Delivery Coupon Discount	High(81.8%)	Medium	Low + Medium	Medium
Delivery Sales per Visit	Medium(53.4%)	Medium	Medium	Medium
Delivery Sales	High(97.4%)	Medium	Low+Medium	Medium
Delivery Visits	High(95.7%)	Medium	Medium	Medium
Non-Kroger Sales	Medium	High(75.6%)	Medium	Medium
Kroger Sales	Medium	High (61.9%)	Medium	Low
Coupon Frequency	High (58.2%)	High (62.9%)	Medium	Medium
Yellow Tag Frequency	High (51.4%)	High (66.8%)	Medium	Medium
n	1349	1408	1886	5357
	Delivery	In store	Pick-up	Assorted



Overview of Clusters

In Store Cluster

- 14% of customers
- 79% of sales were in store
- 96% of customers were loyal.
- Highest spending, visits, and coupon usage of the clusters.

Suggested Goals: Target cluster with in store ads, coupons, and promotions. Determine if Kroger wants to keep customers shopping instore or if they want them shopping online.

Pick-up Cluster

- 19% of customers.
- 53% of sales were pick-up orders.
- 40% of sales were in store.
- 90% of customers were loyal. More likely to use coupons on pick-up orders than in store trips.

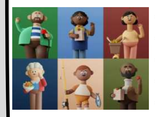
Suggested Goals: Target customers with pick-up ads, coupons, and promotions.

Delivery Cluster

- 13% of customers.
- 54% of sales were delivery orders.
- 38% of sales were in store.
- 85% of customers were loyal.
- High coupon usage.
- More likely to use coupons on delivery orders than in store trips.

Suggested Goals: Target customers with delivery ads, coupons, and promotions.

Assorted Cluster



- 54% of customers.
- 54% of sales were in store.
- 26% of sales were pick-up orders.
- 20% of sales were delivery orders.
- 58% of customers were loyal.
- 25% of customers were opportunity.
- 10% of customers were non-loyal.
- Lowest sales, visits, and coupon usage.

Suggested Goals: Further analyze this cluster by collecting and examining more data on customer behavior and shopping habits. Turn opportunity customers into loyal customers.

Next Steps

- We have identified Kroger customer segmentations.
- We can update our current dataset by adding a new column called "Segmentation."
- Now Kroger can use the updated data to build a classification rule and use this to classify new customers.
- We believe our findings will help Kroger know their customers best!

Acknowledgements

We express our sincere gratitude to 84.51° and The Kroger Company for their invaluable support and provision of data, which made the project possible. Special thank you to our mentors Dr. Dhanuja Kasturiratna and Dr. Lisa Holden and industry partner Mr. Parker Kain for their unwavering guidance and insightful feedback throughout the research process. Additionally, we would like to thank our colleagues and peers for their continuous support and encouragement.